



Inspection to Prediction- Unraveling the link between UT informed drilling induced delamination and failure of carbon fiber thermoset polymers



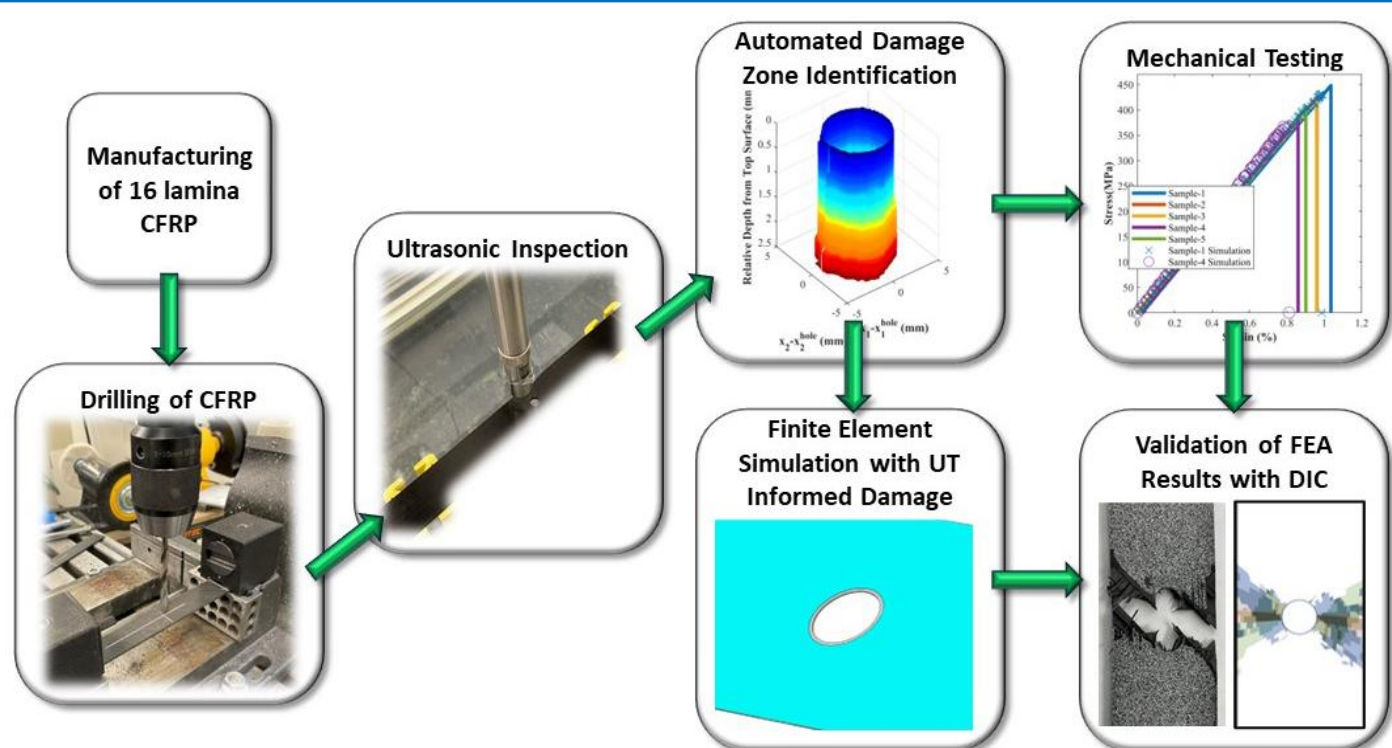
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Introduction

Although drilling in composites is often required to facilitate the mechanical joining between parts, it causes delamination, burrs, and chips around the hole rim when worn-off drill bits (improper tool) or high spindle speed and low feed rates (improper setting) are selected. Drilling-induced damage may cause a premature failure of composites, and method to identify accurately the damage to predict the failure is required. Traditionally, a 2-step approach, predicting damage during drilling on a ply-by-ply basis, followed by structural simulation with applied load is employed which is computationally expensive and require validation on both steps. Objectives: Our study aims to propose a novel framework for:

- Identifying accurately the interlaminar damage (delamination) in composite laminates due to drilling using Ultrasound Scanning system.
- Translating the identified damage into 3D finite element model as delamination in cohesive element,
- Predicting damage progression and failure onset of CFRP laminates using continuum damage mechanics.

Novel Framework



Manufacturing, Drilling process and UT Scanning

Panels of UD carbon/ epoxy T700/250F of 16-laminae with a quasi-isotropic lay-up $[0^\circ/-45^\circ/+45^\circ/90^\circ]_{2S}$ were manufactured using hot press. Samples of 250mm x 25mm were then drilled using 6.25 mm diameter drill bit (135° tip angle) at 1700 rpm to introduce delamination (**Figure-1**).

Drilling-induced delamination was scanned using in-house UT immersion system (15MHz transducer) in x_1 (scan), x_2 (index), and x_3 (focus adjustment) directions (**Figure-2**)



Figure-1. Drilling Process

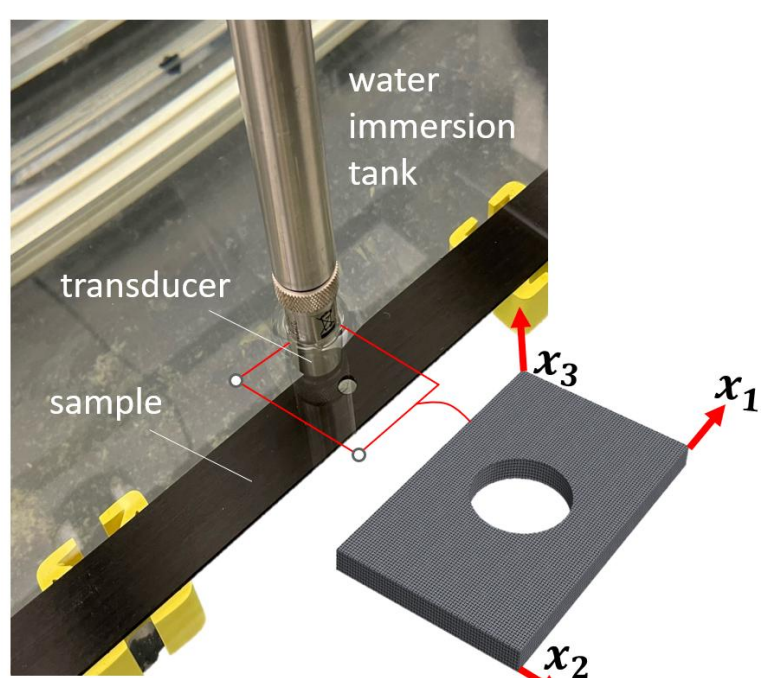


Figure-2. UT Immersion system

Abstract: Along with the decrease brought on by drilling, there is also a further decline in mechanical performance due to delamination if the right drilling parameters are not followed. To correlate the failure of composite to delamination, a novel method by Ultrasound to identify and quantify the drilling induced damage (delamination) in each layer is developed. In contrast to conventional open hole tension test, this study uses the damage information found from Ultrasound analysis to design the finite element model for simulating open hole tension test. A custom built exponential progressive damage modelling approach to account for the matrix, fiber failure is designed while cohesive zone modelling approach takes interface delamination into account. A very good agreement is demonstrated by comparing simulation results to experimental results in terms of stiffness and surface strain field obtained by digital image correlation, damage progression, and failure in composites. By including observed damage features, this UT-informed modeling approach provides a dependable way to evaluate the in-service loading capability of produced composites.

Identification of delamination using UT

At each interface, C-scan energy (**Figure-3**) around the hole are compared to undamaged regions. At same depth, delamination is identified by thresholding waveform's energy relative to undamaged signal resulting in a 3D profile of the damage (**Figure-4**).

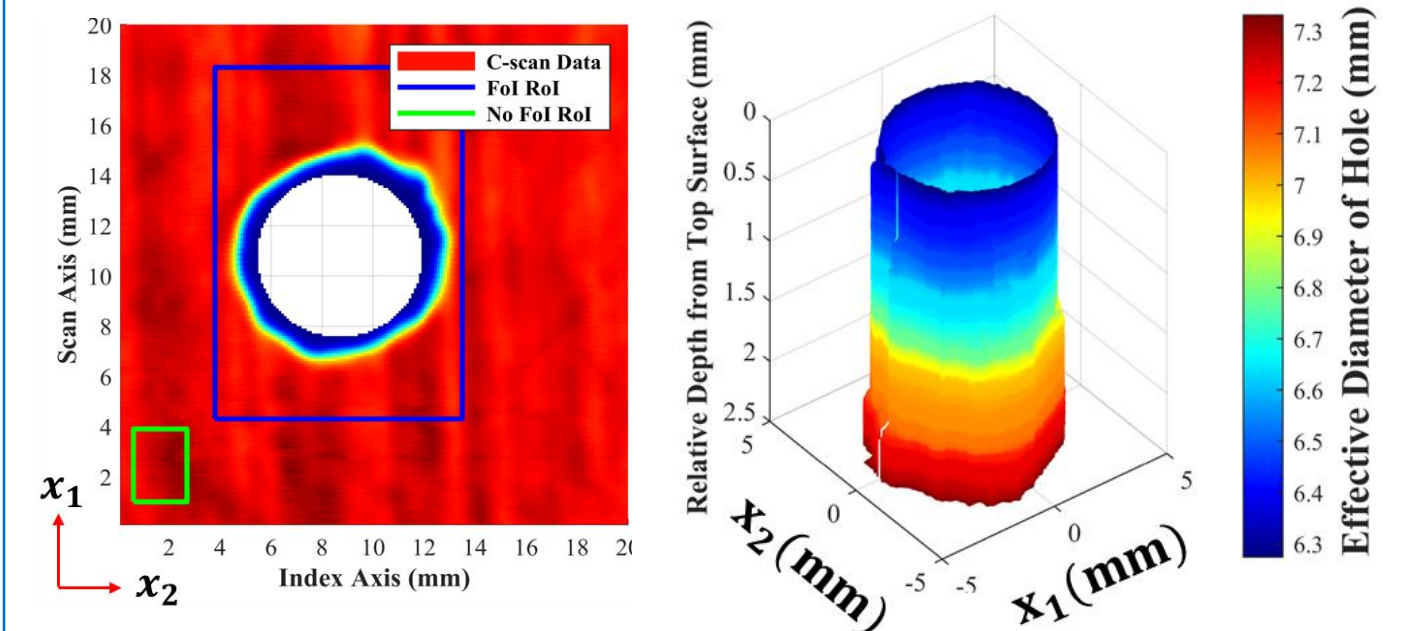


Figure-3. C-Scan

Figure-4. 3D Damage Area

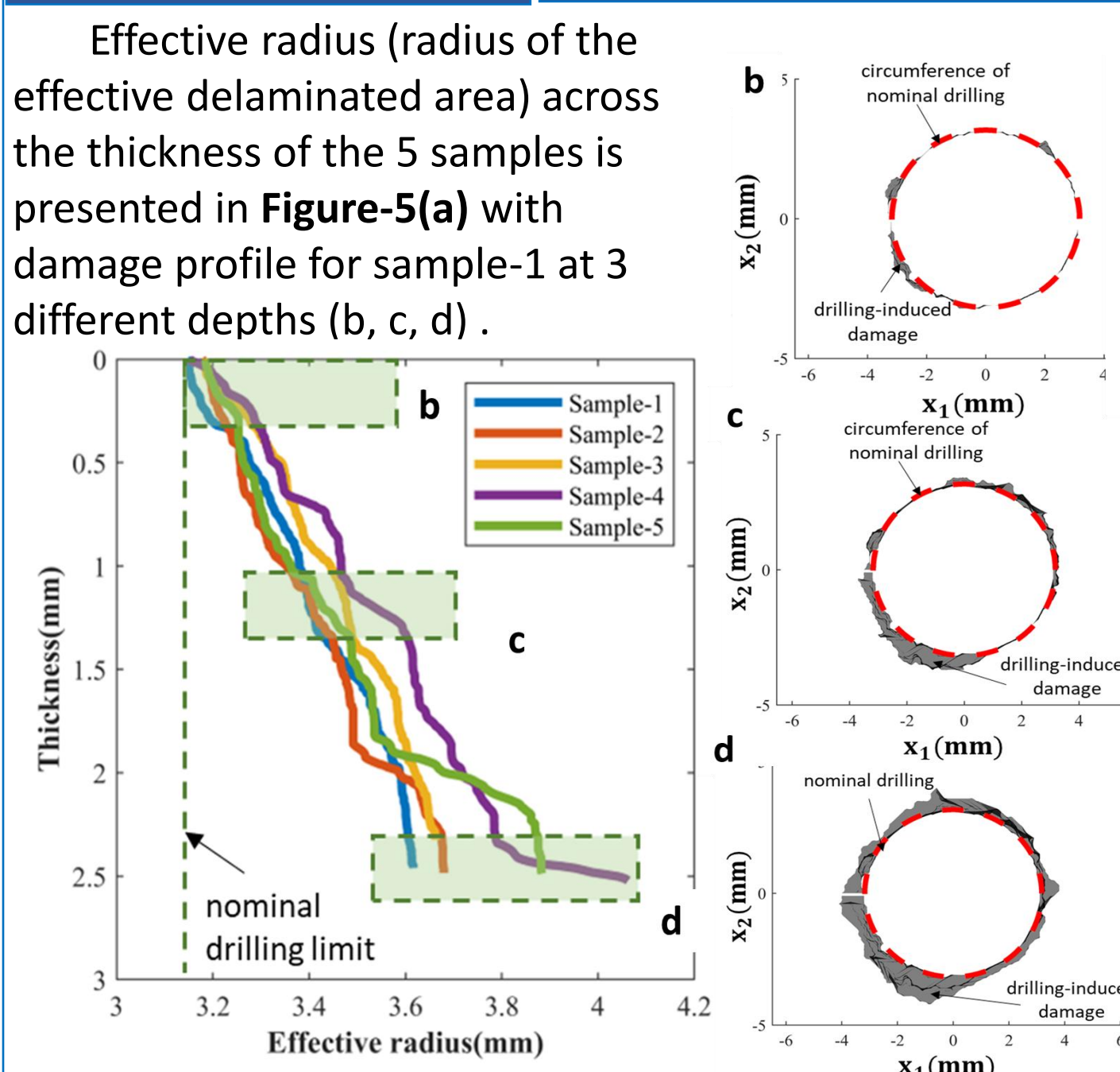


Figure-5. (a) Effective radius variation for 5-samples, Drilling induced damage at 3 different depths (b) 0.1 mm, (c) 1.25 mm, (d) 2.25 mm

Transforming UT data to Finite Element Domain

- Mesoscale model
- VUMAT was employed for 3D ply Failure criteria with exponential progressive damage evolution law
- The effective radius of the delaminated interface area in each ply is defined as the effective radius of the hole of cohesive zone (**Figure-6**)

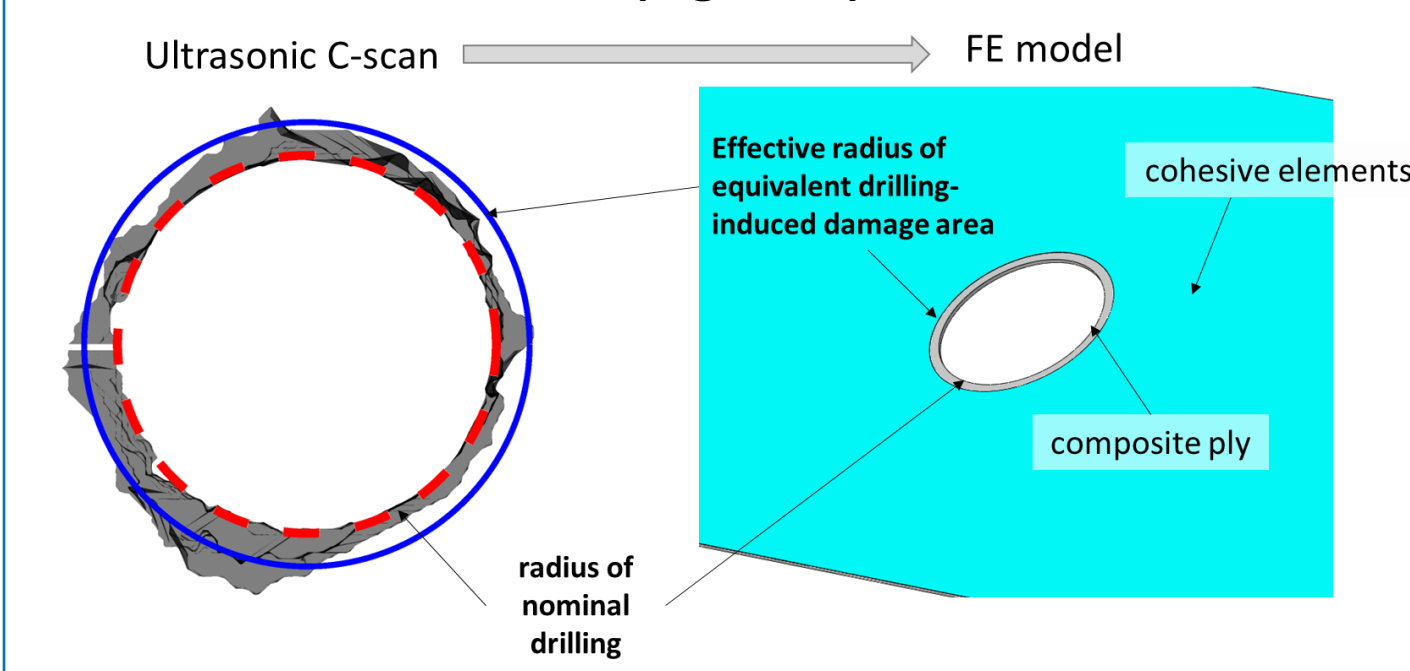


Figure-6. Transforming ultrasonic C-scan data of drilling-induced damage/delamination into FE model

Mechanical response of open hole tension

All five samples exhibit similar effective stiffness at an average of 45.5 ± 2.3 GPa. However, there is variability in failure stress, averaging 416 ± 25 MPa, and failure strain at $0.94\% \pm 0.07\%$ (**Figure-7**). **Figure-8** indicates that there is a monotonic decrease in the failure strain percentage as a function of increasing delamination area.

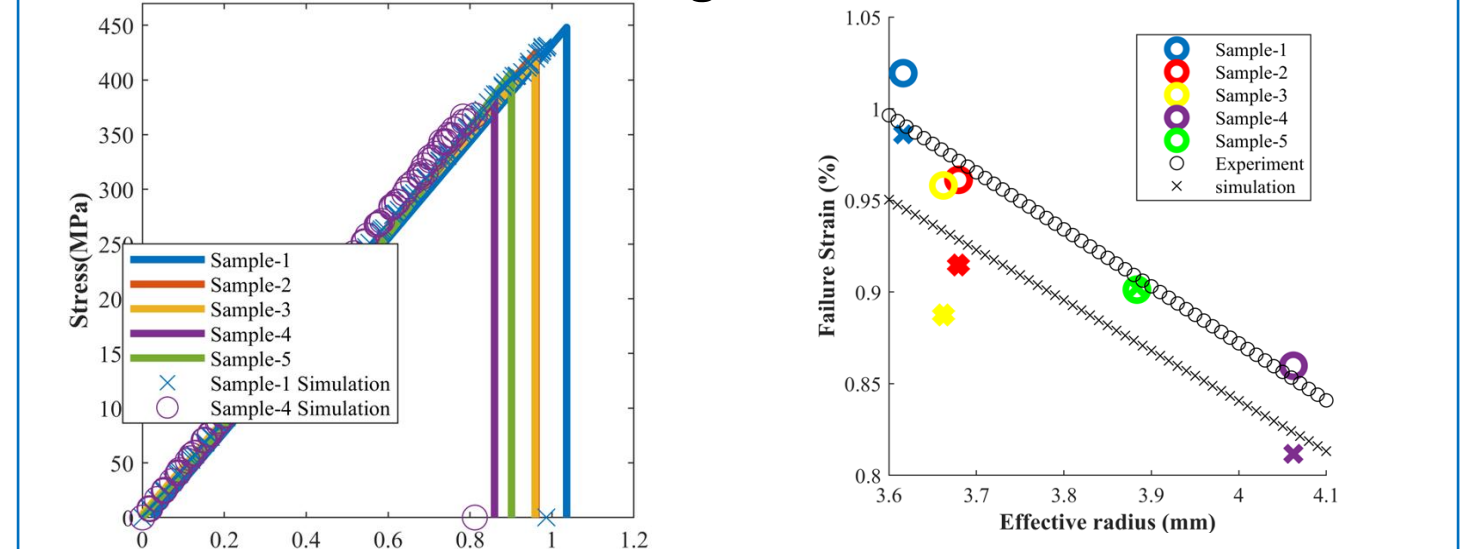


Figure-7. Stress-Strain curves from FEA and experiment

Figure-8. Correlation between failure strain and effective radius

Strain field evolution at the hole rim

The strain field from the experimental digital image correlation is compared to the finite element simulations for Sample-4 shown in **Figure-9**.

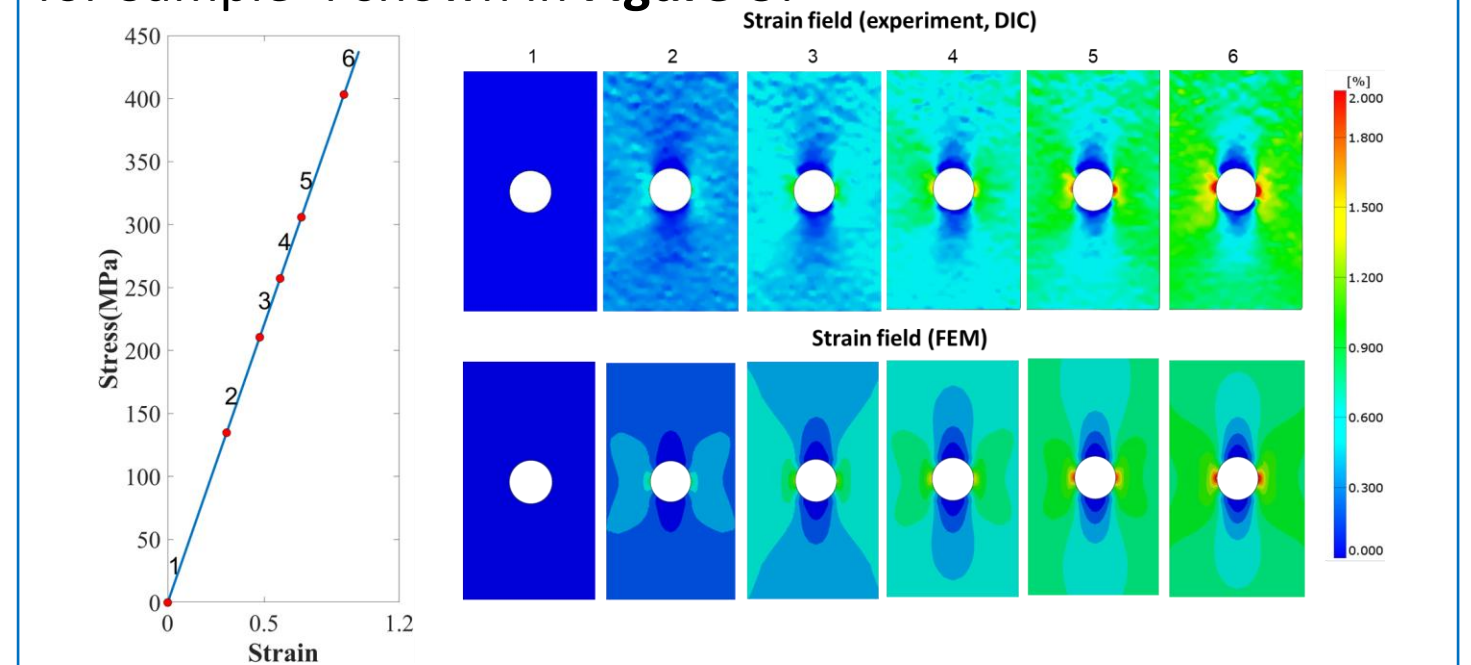


Figure-9. Strain field evolution from DIC and FEA

Progressive damage

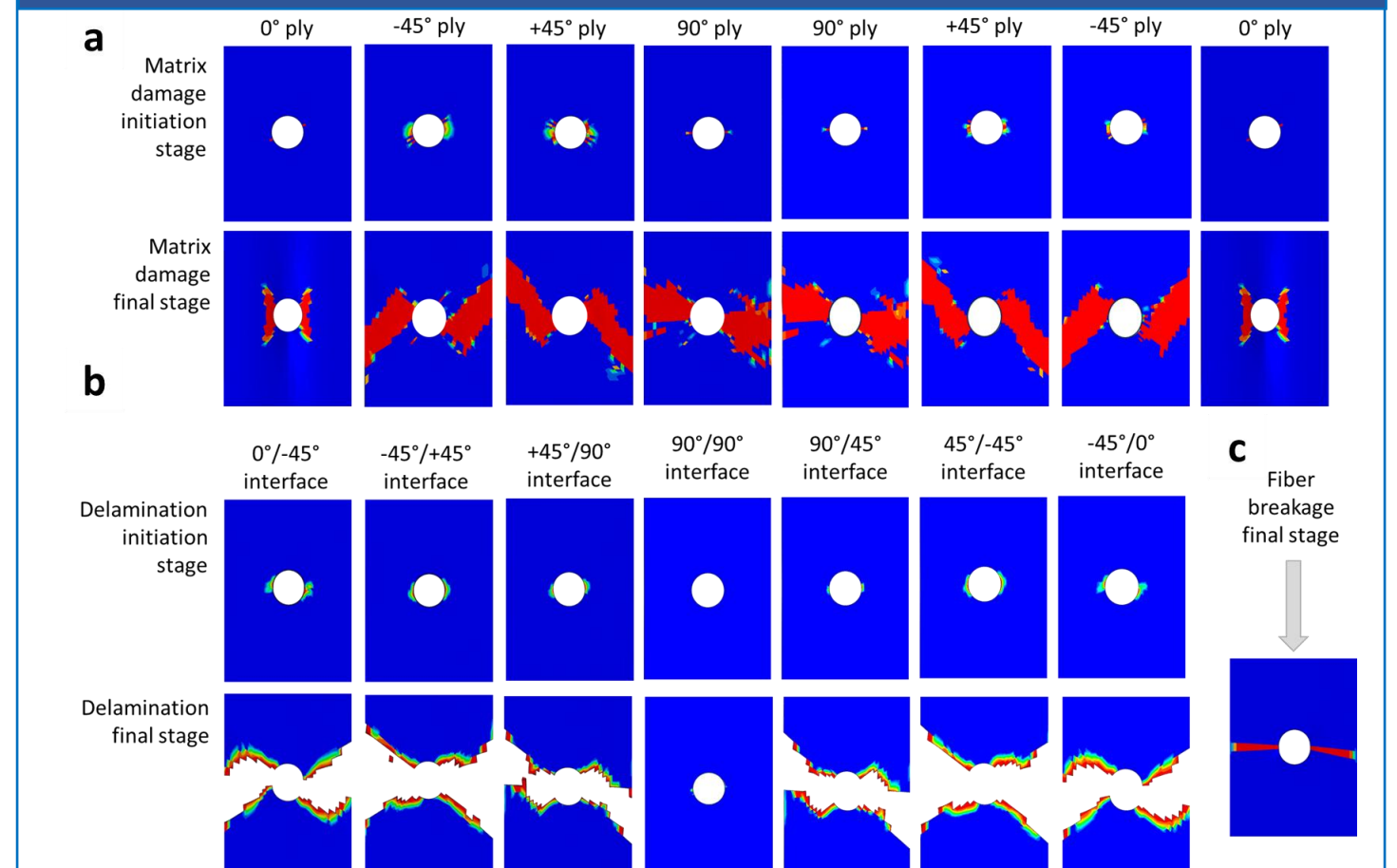


Figure-10. (a) Matrix damage, (b) Interface Delamination, (c) Fiber breakage

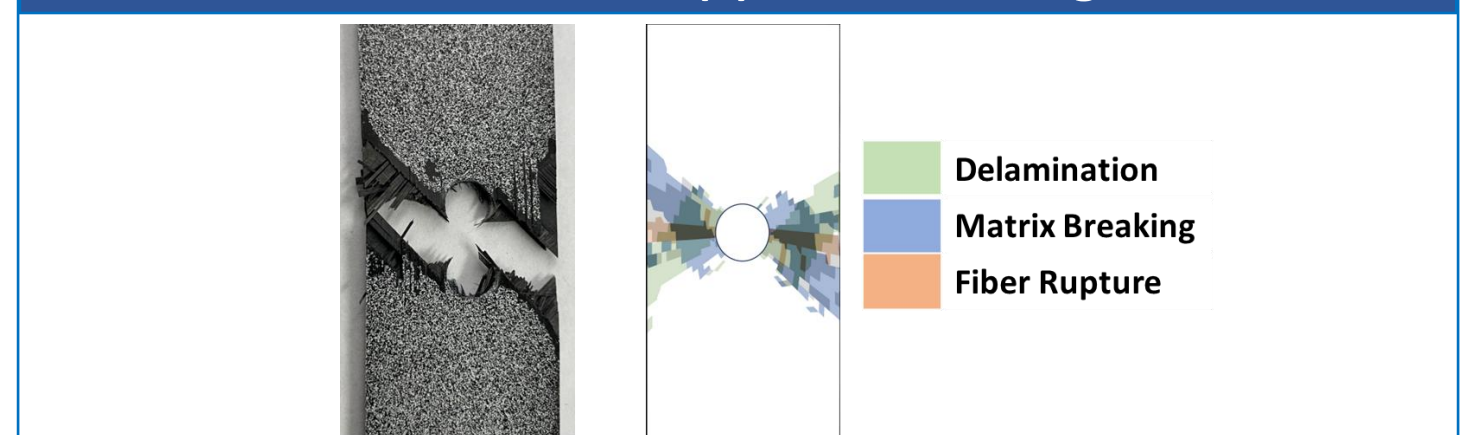


Figure-11. Overall Failure of physical and simulated part

A novel framework to identify interlaminar delamination due to drilling using UT scan, and for translating that into 3D finite element model is developed by designing cohesive element for delamination and ply degradation by continuum damage mechanics. The FE model was able to predict the tensile strength and corresponding damage progression during static loading. This new experimental-numerical framework will be useful for simulating similar cases involving UT-identified damage and composite failure.